

Axel Rodríguez-Pérez

BSC IN GENOMIC SCIENCES

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Education

Bachelors in Science in Genomic Sciences

NATIONAL AUTONOMOUS UNIVERSITY OF MEXICO (UNAM)

Juriquilla, Qro, MX

2021 – 2025

- Relevant Coursework: *Calculus, Statistics, Linear Algebra, Molecular and Cell Biology, Statistical Models in Human Genetics, Human Genomics in Health and Disease, Cancer Genomics, Evolutionary Genomics, Bioinformatics and Statistics, Machine Learning Applications in Genomics.*

Research Experience

Research Intern

LABORATORY OF MOLECULAR CANCER BIOLOGY, CENTER FOR CANCER BIOLOGY, VIB - KU LEUVEN

Leuven, Belgium

August 2024 – June 2025

Supervisors: Jean-Christophe Marine, PhD. & Joanna Pozniak, PhD.

- Analyzed the perivascular niche in melanoma using Molecular Cartography spatial transcriptomics data.
- Designed pipelines for transcript assignment, quality assessment, and robust cell-type annotation.
- Built spatial neighbor graphs and co-occurrence models to study cell-cell interactions.
- Contributed to and adapted Python pipelines for processing multiplexed tissue imaging data (NanoNail and CIVO micro-dosing devices)
- Implemented automated image preprocessing and segmentation workflows using Cellpose and Dask for large-scale datasets.
- Performed spatial analyses and clustering to evaluate treatment responses at single-cell resolution.
- Processed and analyzed bulk RNA-seq data from treated melanoma cell lines using HPC workflows.
- Identified treatment-specific gene signatures and enriched pathways through differential expression and functional analysis.
- Quantified volumetric and surface changes in melanoma cell lines under Palbociclib treatment.
- Applied vector-based models to characterize magnitude and direction of cellular responses.

Research Intern

CANCER GENOMICS AND BIOINFORMATICS LAB, INTERNATIONAL LABORATORY FOR HUMAN GENOME RESEARCH (LIIGH -UNAM)

Juriquilla, Qro, MX

May 2022 – July 2024

Supervisors: Carla Daniela Robles Espinoza, PhD. & Martha Estefanía Vázquez Cruz, PhD.

- Analyzed spatial proteomics data from acral lentiginous melanoma patient samples using GeoMx Digital Spatial Profiler (35 protein markers).
- Developed and benchmarked machine learning models (Random Forest, Logistic Regression) to predict ulceration status from tumor and TME regions.
- Applied correlation filtering, PCA, and upsampling with 10-fold cross-validation (PR-AUC $\approx$ 0.98 in TME samples).
- Identified immune-related markers (e.g., CD8, PD-1) as the strongest predictors of ulceration.
- Documented analyses in R and prepared a technical report with reproducible workflows.
- Gained hands-on training in molecular biology techniques, including nucleic acid extraction, quantification, and agarose gel electrophoresis.
- Cultured mammalian cell lines (NIH 3T3, A375 melanoma): thawing, passaging, contamination detection, and cryopreservation.
- Performed bacterial transformation, plasmid amplification, and purification with quality control.
- Applied functional assays (foci formation, crystal violet staining, scratch, proliferation) to assess oncogenic potential.
- Conducted mammalian cell transfection and antibiotic selection; prepared buffers, reagents, and sterile culture media.

Publications

1. Vazquez-Cruz, M. E., Basurto-Lozada, P., Molina-Aguilar, C., Orozco-Ruiz, S., Van Haastrecht, B., Simonin-Wilmer, I., Martinez-Said, H., Alvarez-Cano, A., Garcia-Ortega, D. Y., Hidalgo-Miranda, A., Hinojosa-Ugarte, D., Ferreira, I., Tavares-De-La-Paz, L. A., Olguin, J. E., Salinas, I., **Rodriguez-Perez, A.**, Martinez-Gomez, J. M., Van Der Zee, I., Grimes, D. R., . . . Robles-Espinoza, C. D. (2025). The microenvironment of ulcerated acral melanoma is characterised by an inflammatory milieu and an enhanced humoral immune response. *medRxiv (Cold Spring Harbor Laboratory)*. <https://doi.org/10.1101/2025.05.05.25325616>

Presentations & Talks

Machine Learning models for ulceration prediction in mexican melanoma cohort

*International Laboratory for Human
Genome Research (LIIGH-UNAM)*

MINI-SYMPOSIUM: ACRAL MELANOMA RESEARCH IN LATIN AMERICA

May 2024

- Presented research on applying machine learning (Random Forest, Logistic Regression, LASSO) to spatial proteomics data from Mexican acral melanoma patients.
- Discussed methodology (data preprocessing, feature selection, PCA, upsampling) and evaluation metrics (PR-AUC).
- Discussed where such models could fit into melanoma prognosis in the clinic.

Teaching & Mentorship Experience

Invited Speaker

Remotely

5TH SUMMER SCHOOL: INTRODUCTION TO GENOMIC SCIENCES, LIIGH-UNAM

July 2025

Talk title: *Genes, code, and decisions: lessons on starting a scientific career.*

- Delivered an interactive talk as part of the “My Science and My Life” series.
- Shared personal trajectory and lessons for early-career students, combining genomics, coding, and career decision-making.

Guest Instructor

Remotely

UNDERGRADUATE PROGRAM IN GENOMIC SCIENCES, UNAM

November 2024

- Assisted in teaching “*Introduction to Single-Cell and Spatial Transcriptomics*” lecture for third-semester undergraduate students.
- Clarified core concepts and answered student questions.

Achievements & Funding

- UNAM High Academic Performance Scholarship 2023-2024.
- UNAM High Academic Performance Scholarship 2022-2023.
- UNAM High Academic Performance Scholarship 2021-2022.
- UNAM Financial Support Scholarship 2021.

Courses

Hands-on introduction to targeted spatial transcriptomics data analysis

VIB TRAINING

Leuven, Belgium

December 2024

Machine Learning and Deep Learning Workshop

VIB TRAINING

Ghent, Belgium

November 2024

Nextflow for reproducible and automated data analysis

VIB TRAINING

Leuven, Belgium

November 2024

Introduction to NGS analysis

VIB TRAINING

Leuven, Belgium

October 2024

Docker and Apptainer (Singularity) for reproducible and automated data analysis

VIB TRAINING

Leuven, Belgium

October 2024

Technical Skills & Interests

Programming Languages

R, PYTHON, BASH, SQL, MATLAB, GIT

Bioinformatic skills

GITHUB, HIGH PERFORMANCE COMPUTING (HPC), CONDA, DOCKER, NEXTFLOW, RMARKDOWN, QUARTO, JUPYTER NOTEBOOKS

Data Analysis and Visualization

TIDYVERSE, GGLOT2, PLOTLY, SEABORN, MATPLOTLIB, PANDAS, NUMPY

Omics data types

BULK RNA-SEQ, SINGLE-CELL RNA-SEQ, SPATIAL TRANSCRIPTOMICS, PROTEOMICS, GENOME WIDE ASSOCIATION STUDIES (GWAS), NEXT-GENERATION SEQUENCING (NGS)

Laboratory skills

DNA/RNA EXTRACTION AND QUANTIFICATION, MAMMALIAN CELL CULTURE, BACTERIAL TRANSFORMATION, PLASMID PREP, LIPOFECTAMINE TRANSFECTION, FUNCTIONAL ASSAYS

Languages

SPANISH (NATIVE), ENGLISH (PROFICIENT)

Interests

COMPUTATIONAL BIOLOGY AND CANCER GENOMICS

I work on computational biology and cancer genomics, mostly with spatial and single-cell omics. My aim is to apply statistical and machine learning methods to cellular heterogeneity and to how tumors and their microenvironment respond to therapy. I also build reproducible multi-omics workflows that scale on high-performance computing systems.